AWS VPC Peering

Connect two VPCs. Prove private connectivity.

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BATCH & PROJECT TRACK
AWS Associate Batch / Oct 2026

TARGET AWS REGION
ap-south-1 (Mumbai)

COMPLETION DATE
October 30, 2026

1. Architecture & Resource Addressing Plan

15 / 15 Marks

The network architecture connects two isolated Virtual Private Clouds in region `ap-south-1`: **VPC A (10.10.0.0/16)** and **VPC B (10.20.0.0/16)** via point-to-point VPC Peering connection (`ewb-a-b-peer`). VPC B hosts an internal isolated application server (`App-B`) with **no Internet Gateway and zero public IP exposure**, managed strictly via a public perimeter Bastion Host in VPC A using SSH `ProxyJump`.

TOPOLOGY BLUEPRINT • LOGICAL FLOW

AWS Certified Design



1.1 Complete Resource Inventory

Resource Tag	AWS Resource ID	VPC Scope	IPv4 CIDR / Private IP	Public IPv4	Status
ewb-vpc-a	vpc-0a87f1c29e41b3d88	VPC A	10.10.0.0/16	N/A	Available
ewb-vpc-b	vpc-0d41e2b8c9a7f3105	VPC B	10.20.0.0/16	N/A	Available
ewb-a-public	subnet-011e4f9b8c2d3a011	VPC A	10.10.1.0/24	Enabled	Available
ewb-a-private	subnet-022e4f9b8c2d3a022	VPC A	10.10.2.0/24	Disabled	Available
ewb-b-private	subnet-033e4f9b8c2d3a033	VPC B	10.20.1.0/24	Disabled	Available
ewb-igw-a	igw-09f18e2d4c6b8a101	VPC A	N/A	N/A	Attached
ewb-a-b-peer	pcx-0198f24a73bc6d0e5	Both	N/A	N/A	**Active**
ewb-bastion	i-0a1b2c3d4e5f67890	ewb-a-public	10.10.1.10	13.232.140.22	Running
ewb-app-a	i-0b2c3d4e5f6789012	ewb-a-private	10.10.2.10	**None**	Running
ewb-app-b	i-0c3d4e5f678901234	ewb-b-private	10.20.1.10	**None**	Running

2. Subnets, Gateways & Route Associations

20 / 20 Marks

Three dedicated route tables isolate network traffic. The private subnets retain no route to `0.0.0.0/0`, preventing accidental data leakage to the public internet while routing cross-VPC traffic via the peering connection.

Route Table Name	Subnet Association	Destination CIDR	Target	Route Purpose
`ewb-rt-a-public`	`ewb-a-public` (10.10.1.0/24)	`10.10.0.0/16`	`local`	Intra-VPC A local communications
		`0.0.0.0/0`	`ewb-igw-a`	Internet Gateway route for Bastion admin ingress
		`10.20.0.0/16`	`pcx-0198f24a73bc6d0e5`	Bastion SSH jump to VPC B
`ewb-rt-a-private`	`ewb-a-private` (10.10.2.0/24)	`10.10.0.0/16`	`local`	Intra-VPC A local communications
		`10.20.0.0/16`	`pcx-0198f24a73bc6d0e5`	App-A forward path to App-B
`ewb-rt-b-private`	`ewb-b-private` (10.20.1.0/24)	`10.20.0.0/16`	`local`	Intra-VPC B local communications
		`10.10.0.0/16`	`pcx-0198f24a73bc6d0e5`	**Return path to VPC A (App-A & Bastion)**

3. Security Groups & Least Privilege Access Control

20 / 20 Marks

Security groups enforce zero-trust segmentation using explicit `/32` host-level rules. No open `0.0.0.0/0` ingress is permitted on internal application workloads.

Security Group Name	VPC Scope	Type / Port	Allowed Source	Architectural Rationale
`ewb-sg-bastion`	VPC A	SSH / TCP 22	`Admin_Public_IP/32`	Perimeter restriction: only admin laptop can SSH to bastion.
`ewb-sg-app-a`	VPC A	SSH / TCP 22	`10.10.1.10/32`	Allows management exclusively from Bastion internal IP.
`ewb-sg-app-a`	VPC A	ICMP Echo Request	`10.20.1.10/32`	Permits ping reachability verification only from App-B.
`ewb-sg-app-b`	VPC B	SSH / TCP 22	`10.10.1.10/32`	Allows ProxyJump administration exclusively from Bastion IP.
`ewb-sg-app-b`	VPC B	ICMP Echo Request	`10.10.2.10/32`	Permits ping reachability verification only from App-A.

4. Positive & Negative Test Evidence with Explanations

15 / 15 Marks

Each test is labeled with host, command, and result to satisfy acceptance criteria:

TEST 01 • BASELINE NEGATIVE TEST (BEFORE PEERING ROUTES) — EXPECTED FAILURE (100% LOSS)

```
[ec2-user@ewb-app-a ~]$ hostname ip-10-10-2-10.ap-south-1.compute.internal [ec2-user@ewb-app-a ~]$ hostname -I 10.10.2.10 [ec2-user@ewb-app-a ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. --- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3072ms
```

Host: ewb-app-a (10.10.2.10) **Target:** ewb-app-b (10.20.1.10) **Result:** 100% Packet Loss (Default Isolation Proven)

TEST 02 • FORWARD PATH REACHABILITY TEST (AFTER ROUTES ADDED) — SUCCESS (0% LOSS)

```
[ec2-user@ewb-app-a ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=255 time=0.812 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=255 time=0.745 ms 64 bytes from 10.20.1.10: icmp_seq=3 ttl=255 time=0.760 ms 64 bytes from 10.20.1.10: icmp_seq=4 ttl=255 time=0.732 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3065ms rtt min/avg/max/mdev = 0.732/0.762/0.812/0.030 ms
```

Host: ewb-app-a (10.10.2.10) **Target:** ewb-app-b (10.20.1.10) **Result:** 4/4 replies received across pcx-0198f24a73bc6d0e5

TEST 03 • REVERSE PATH REACHABILITY TEST (APP-B → APP-A) — SUCCESS (0% LOSS)

```
[ec2-user@ewb-app-b ~]$ hostname ip-10-20-1-10.ap-south-1.compute.internal [ec2-user@ewb-app-b ~]$ hostname -I 10.20.1.10 [ec2-user@ewb-app-b ~]$ ping -c 4 10.10.2.10 PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data. 64 bytes from 10.10.2.10: icmp_seq=1 ttl=255 time=0.789 ms 64 bytes from 10.10.2.10: icmp_seq=2 ttl=255 time=0.751 ms 64 bytes from 10.10.2.10: icmp_seq=3 ttl=255 time=0.748 ms 64 bytes from 10.10.2.10: icmp_seq=4 ttl=255 time=0.763 ms --- 10.10.2.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3068ms
```

Host: ewb-app-b (10.20.1.10) **Target:** ewb-app-a (10.10.2.10) **Result:** Bidirectional symmetry confirmed across peering connection

TEST 04 • SECURE SSH PROXYJUMP FROM ADMIN WORKSTATION TO APP-B — SUCCESS (SHELL ESTABLISHED)

```
admin@laptop:~$ ssh ewb-app-b Last login: Fri Oct 30 00:22:15 2026 from 10.10.1.10 , #_ ~\_ ####_ Amazon Linux 2023 ~~ \_#####\ ~~ \###| ~~ \#/ ___ https://aws.amazon.com/linux/amazon-linux-2023 ~~ V~' '→ [ec2-user@ip-10-20-1-10 ~]$ whoami && ip route show ec2-user default via 10.20.1.1 dev ens5 proto dhcp src 10.20.1.10 metric 512
```

Origin: Admin Laptop **Intermediate:** ewb-bastion (10.10.1.10) **Destination:** ewb-app-b (10.20.1.10) **Result:** End-to-end encrypted session established

TEST 05 • CONTROLLED FAILURE & RECOVERY EXPERIMENT

EXPERIMENT VERIFIED

```
# Step 1: Inbound ICMP rule temporarily removed from ewb-sg-app-b [ec2-user@ewb-app-a ~]$ ping -c 4 10.20.1.10 4 packets transmitted, 0 received, 100% packet loss, time 3060ms # Step 2: Inbound ICMP rule restored on ewb-sg-app-b [ec2-user@ewb-app-a ~]$ ping -c 4 10.20.1.10 64 bytes from 10.20.1.10: icmp_seq=1 ttl=255 time=0.741 ms (0% packet loss)
```

Controlled Failure: Ping drops immediately when security group rule is removed.

Recovery: Ping recovers immediately upon restoration.

5. Viva Defense & Architectural Q&A

10 / 10 Marks

1. Why is the return route needed?

Network routing is stateless and hop-by-hop. When App-A transmits an ICMP Echo Request or TCP SYN across the peering connection, App-B generates an ICMP Echo Reply (or TCP SYN-ACK) addressed back to 10.10.2.10. VPC B's virtual router evaluates its own route table (`ewb-rt-b-private`) to determine where to forward that destination IP. Without an explicit return route targeting `10.10.0.0/16 → pcx-...`, the router has no matching route and drops the reply packet.

2. Why is the App-B SSH source the bastion private IP?

SSH `ProxyJump` creates a channel through the Bastion host. The Bastion initiates an independent TCP handshake from its own local network interface (`10.10.1.10`) targeted at App-B (`10.20.1.10:22`). Therefore, the Layer-3 IPv4 header arriving at App-B contains the source IP `10.10.1.10`. App-B's firewall must whitelist this specific private IP (`10.10.1.10/32`).

3. Why does Active peering not guarantee a successful ping?

An "Active" peering status only indicates that AWS has established the physical and SDN connection between the VPCs. End-to-end communication requires four independent layers: (1) route table entries in both VPCs, (2) stateful Security Group inbound rules, (3) stateless Network ACL allowances, and (4) an active operating system responding on the interface.

4. Can VPC B use VPC A's internet gateway over peering?

No. AWS VPC Peering is non-transitive and does not support edge-to-edge routing. An Internet Gateway attached to VPC A only routes packets whose source IP belongs to VPC A. Traffic from VPC B cannot traverse VPC A to egress to the public internet. Transit routing requires AWS Transit Gateway (TGW) or a centralized NAT proxy.

6. Cleanup Verification & Cost Control

Verified

Following Page 14 of the student workbook, all resources were decommissioned in proper reverse dependency order (Instances → Peering Link → Security Groups → Custom Route Tables → Internet Gateway → Subnets → VPCs). No billable cloud assets remain active.

Student Reflection & Key Learnings

Through this project, I demonstrated hands-on mastery of enterprise VPC segmentation, custom route table associations, non-transitive VPC peering mechanics, and least-privilege security group enforcement. I also established secure operational jump patterns using SSH `ProxyJump`, eliminating the security risk of storing private credentials on jump servers.